measure the load current onboard.

Variable-output power supply. To convert the PSU into a variable power supply, use an LM338K or LM317 adjustable voltage regulator IC as shown in Fig. 6. Connect +12V and -12V supplies from the PSU to this circuit so that you get 1.2V to +23V maximum output using VR1 potmeter. But the current output will reduce to 500mA because the output current of -12V output is 500mA only.

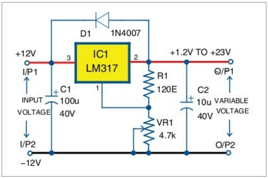

Fig. 6: Adjustable regulated power supply circuit

Construction and testing

An actual-size PCB layout for variable power supply is shown in Fig. 7 and its components layout in Fig. 8. Some tips for testing the power supply are:

1. Some newer PSUs come with a current-sense wire. If the wire is grey in colour, connect it with +3.3V. If it is pink, connect it with +5V supply.
2. Make sure that you are using an ATX power supply. If it is an older one, it will have different colour codes for the wires. If you don't have the actual colour code data, don't try anything.
3. If the LEDs do not light up, check these for correct polarity.
4. The cooling fan in the PSU may make noise. As the PSU is not powering a load continuously, you can limit the fan speed to reduce the noise. You can do so by reducing the supply voltage from 12V to 5V. Consequently, if the PSU heats up abnormally, reconnect the fan with 12V supply.
5. It is recommended to connect a separate fuse for each output.

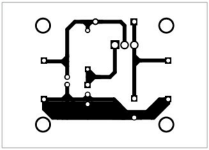

Fig. 7: Actual-size PCB layout of the variable power supply circuit

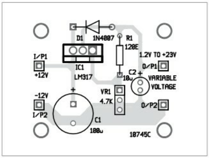

Fig. 8: Components layout for the PCB

Advantages of PSU

1. Very effective short-circuit protection
2. Thermal protection
3. Earth fault protection
4. Compact size with respect to voltage and current output ratings
5. Very cheap compared to conventional bench power supplies

Disadvantages

1. Complex circuit, difficult to repair without adequate knowledge
2. SMD soldering station may be required for repair EFY

Jaywardhan Pandit is an electronics hobbyist. His interests include embedded system and PCB designing